Lecture 17

09.12.25

Remark (Neumann b.c.) consider $-u''(x) = f(x)$ in $(0,1)$

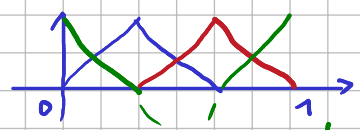
$$u'(0) = g_0 \,,\; u'(1) = g_1$$

$$\Rightarrow \int_0^1 f(x)v(x)\,dx = -\int_0^1 u''(x)v(x)\,dx = \int_0^1 u'(x)v'(x)\,dx - \left[u'(x)v(x)\right]_0^1$$

$$\Rightarrow \text{weak form} \quad \int_0^1 u'(x)v'(x)\,dx = \int_0^1 f(x)v(x)\,dx + g_1 v(1) - g_0 v(0)$$

All we need is for u and v to be once weakly differentiable!

In particular, the finite-dimensional trial space V_h does <u>not</u> need to incorporate boundary conditions!



can be used as well!

Finite element analysis

Goal: analyze the variational problem

$$\text{Find } u \in V \text{ s.t. } a(u,v) = F(v) \text{ for all } v \in V \qquad (VP)$$

Assumption Let $\Omega \subset \mathbb{R}^d$ be a bounded polygonal domain and set $V = H_0^1(\Omega)$ (for Dirichlet boundary conditions) with norm $\|u\|_V = \|u\|_{H^1(\Omega)}$. Assume further that $a: V \times V \to \mathbb{R}$, $F: V \to \mathbb{R}$ satisfy

a) There exists an $\alpha \in (0,\infty)$ s.t. $|a(v_1,v_2)| \le \alpha \|v_1\|_V \cdot \|v_2\|_V \;\; \forall v_1, v_2 \in H_0^1(\Omega)$.

Then we call a continuous / bounded bilinear form.

b) There exists a $\beta \in (0,\infty)$ s.t. $a(v,v) \ge \beta \|v\|_V^2 \;\; \forall v \in H_0^1(\Omega)$.

Then we call a V-elliptic / coercive.

c) There exists a $\gamma \in (0,\infty)$ s.t. $|F(v)| \le \gamma \|v\|_V \;\; \forall v \in H_0^1(\Omega)$.

Then we call F continuous / bounded linear form.

Theorem (Lax-Milgram) Under the previous assumptions there exists a unique solution of the variational problem (VP) and we have the estimate

$$\|u\|_V \le \frac{\gamma}{\beta}.$$

Note: a is not required to be symmetric.

Recall the <u>Cauchy-Schwarz</u> inequality: Z a Hilbert space with inner product $\langle \cdot,\cdot \rangle_Z$ and induced norm $\|z\|_Z = (\langle z,z\rangle)^{1/2} \Rightarrow |\langle z_1, z_2\rangle_Z| \le \|z_1\|_Z \cdot \|z_2\|_Z$

<u>Comparison with linear algebra problem $Ax = b$</u> $\Leftrightarrow$ $\underbrace{y^T A x}_{a(x,y)} = \underbrace{y^T b}_{F(y)}$

- bilinear form $a(x,y) = y^T A x$

continuity: $|a(x,y)| = |\langle y, Ax\rangle_{\mathbb{R}^n}| \overset{\text{Cauchy Schwarz}}{\le} \|y\| \cdot \|Ax\| \le \|y\| \cdot \underbrace{|||A|||}_{\alpha \in (0,\infty)} \cdot \|x\|$

coercivity: $a(x,x) = x^T A x \overset{!}{\ge} \beta \|x\|^2 \Rightarrow \beta \|x\|^2 \le \langle x, Ax\rangle \le \|x\| \cdot \|Ax\|$ $(*)$

means here: $a(x,x) \ge 0$ and "$=0$" if and only if $x = 0$

$\Leftrightarrow$ A is supposed to be positive definite $\Rightarrow A^{-1}$ exists!

continuity: $|F(y)| = |y^T b| = |\langle y, b\rangle_{\mathbb{R}^n}| \overset{\text{Cauchy Schw}}{\le} \|y\| \cdot \underbrace{\|b\|}_{\gamma}$

$\Rightarrow$ $x = A^{-1} b$ (possible, since A pos. def.) and $\|x\| \le \|A^{-1} b\| \le |||A^{-1}||| \cdot \underbrace{\|b\|}_{\gamma}$

In particular: $|||A^{-1}||| = \sup\limits_{z \ne 0} \dfrac{\|A^{-1} z\|}{\|z\|} = \sup\limits_{\substack{z = Aw \\ w \ne 0}} \dfrac{\|A^{-1}(Aw)\|}{\|Aw\|} = \dfrac{\|w\|}{\|Aw\|} \overset{(*)}{\le} \dfrac{1}{\beta}$

Conclusion: Lax-Milgram is consistent with what we already know from LA

<u>Example</u> (general elliptic operators) $\Omega \subset \mathbb{R}^d$ as above and the operator

$$Lu(x) = -\nabla \cdot (a(x) \nabla u(x)) + c(x) u(x) \qquad (+ \text{ Dirichlet b.c.})$$

where $a, c : \Omega \to \mathbb{R}$ are bounded and smooth functions with

$$a(x) \ge a_0 > 0, \quad c(x) \ge c_0 \ge 0 \qquad \text{for all } x \in \Omega$$

$\Rightarrow$ bilinear form $a(v_1, v_2) = \int_\Omega a(x) \nabla v_1(x) \cdot \nabla v_2(x) + c(x) v_1(x) v_2(x)\, dx$

Note: $a(v_1, v_2) = a(v_2, v_1)$, i.e., a is symmetric

Assumptions satisfied?

- continuity of a:

$$|a(v_1, v_2)| \le \int_\Omega |a(x)| \cdot |\underbrace{\nabla v_1(x) \cdot \nabla_2(x)}_{\langle \nabla v_1(x), \nabla v_2(x)\rangle_{\mathbb{R}^n}}| + |c(x)| \cdot |v_1(x)| \cdot |v_2(x)|\, dx$$

$$\overset{\text{Cauchy-Schwarz for } \langle\cdot,\cdot\rangle_{\mathbb{R}^n}}{\le} \int_\Omega |a(x)| \cdot \|\nabla v_1(x)\| \cdot \|\nabla v_2(x)\|\, dx + \int_\Omega |c(x)| \cdot |v_1(x)| \cdot |v_2(x)|\, dx$$

$$\le \|a\|_\infty \int_\Omega \|\nabla v_1(x)\| \cdot \|\nabla v_2(x)\|\, dx + \|c\|_\infty \int_\Omega |v_1(x)| \cdot |v_2(x)|\, dx$$

where $\|a\|_\infty = \sup_{x\in\Omega} |a(x)|$, $\|c\|_\infty = \sup_{x\in\Omega} |c(x)|$.

Using Cauchy-Schwarz for $\langle f, g\rangle_{L^2(\Omega)} = \int_\Omega f(x)\, g(x)\, dx$ we obtain

$$|a(v_1,v_2)| \leq \|a\|_\infty \|\nabla v_1\|_{L^2(\Omega)} \|\nabla v_2\|_{L^2(\Omega)} + \|c\|_\infty \|v_1\|_{L^2(\Omega)} \|v_2\|_{L^2(\Omega)} \qquad (**)$$

Since $\|v\|_V = \|v\|_{H^1(\Omega)} = \left(\|v\|^2_{L^2(\Omega)} + \|\nabla v\|^2_{L^2(\Omega)} \right)^{1/2}$, it holds that

$$\|\nabla v\|_{L^2(\Omega)} \leq \|v\|_V \quad \text{and} \quad \|v\|_{L^2(\Omega)} \leq \|v\|_V$$

$(**)$ then implies

$$|a(v_1,v_2)| \leq \left(\|a\|_\infty + \|c\|_\infty \right) \|v_1\|_V \|v_2\|_V \quad \rightarrow \text{ continuity of } a \checkmark$$